AKASH S

AI/ML Engineer | Agentic AI · RAG · LLMs · Computer Vision

+91-8610340958 | akash2612005@gmail.com | Akash-2615 | Akash S | Portfolio

Experience

- **Deep Learning Intern, INTRAINZ.** *March 2025 - May 2025*
  - Built a **deepfake video detection system** using a hybrid **3D-CNN + ResNet + BiLSTM** model in **PyTorch** with **OpenCV** frame extraction, reaching **95.4% accuracy** (94.7% precision, 96.2% recall) on a custom frame dataset.
  - Deployed a **Streamlit** app for real-time detection from uploads or webcam, with frame-wise predictions and **Gemini API**-generated explanations of each verdict.
- **Machine Learning & Data Science Intern, NETIX.AI.** *June 2025 - November 2025*
  - Built an **Automated Data Quality Reporting (DQR)** pipeline using **Python, FastAPI, Isolation Forest and Autoencoders** for schema validation, outlier detection and drift monitoring across IoT and energy datasets, **automating 95% of quality checks** and generating **Power BI** reports, leaving only final verification to a human reviewer.
  - Developed an **AI-Driven Smart Building Analytics** system using **Liquid Neural Networks, Neural ODEs and AR feedback** on MQTT-based IoT streams, achieving **98% accuracy on unseen data** versus **92% for an LSTM baseline**, with real-time temperature/CO_2 forecasting, predictive HVAC control and self-retraining.
  - Created a **Multimodal RAG Chatbot** using **FAISS + BM25 hybrid retrieval and Gemini**, ingesting PDF, image, video and CSV via OCR/Whisper, with citation mapping, PDF bounding-box highlights and blockchain-based document verification across **300+ documents**.

Technical Skills

Languages: Python, SQL.
ML & Data Libraries: Scikit-learn, Pandas, NumPy, Hugging Face Transformers, OpenCV, SHAP, Matplotlib, Seaborn.
Frameworks: PyTorch, TensorFlow, LangChain, LangGraph, FastAPI, Flask, React, Celery.
GenAI & Agentic AI: RAG, Multi-Agent Systems, FAISS, Qdrant, Cohere Rerank.
Cloud, Databases & DevOps: AWS (EC2), Docker, Git/GitHub, PostgreSQL, MongoDB, Redis, Supabase.
Soft Skills: Leadership, Creativity, Team Coordination, Adaptability.

Projects

- **ArchitectAI: Multi-Agent Software Architecture Debate Engine** *2026*
  - Tools & Technologies: Python, FastAPI, **LangChain**, asyncio, Anthropic Claude API.
  - Designed a multi-agent AI system with **19 collaborating agents** across **7 structured phases** to evaluate and stress-test software architecture options, automating proposal generation, critique, validation, voting, and final synthesis.
  - Built the agent orchestration and backend pipeline to transform 19 agent perspectives into a unified architecture proposal, generating structured architecture documents with risk analysis, cost considerations, and diagrams, significantly reducing the manual effort involved in architecture planning.
- **SENTINEX: Physics-ML Hybrid Diesel Engine Leak Detection System** *2026*
  - Tools & Technologies: Python, FastAPI, React, PyTorch, SHAP, Docker, AWS EC2, Supabase.
  - Built a 7-stage physics-ML pipeline for diesel engine leak detection, monitoring **42 sensors** across 7 air-path locations with **99.95% accuracy** and sub-1-second detection latency. Integrated ML models, **SHAP explainability**, and sensor fault isolation.
  - Developed a **3D Digital Twin** and ARIA AI copilot for real-time monitoring, visualization, and diagnosis. Deployed the system on **AWS EC2** with support for **federated learning** across multiple test facilities.
- **Forensic Financial Intelligence Suite** *2026*
  - Tools & Technologies: Python, FastAPI, LangChain, **Qdrant**, PostgreSQL, **Redis**, Celery, React, TypeScript.
  - Engineered a **LangChain**-based RAG system for analyzing SEC filings, featuring dual-lane PDF ingestion and Cohere Rerank to improve retrieval quality. Built **automated document processing** and retrieval workflows for efficient financial analysis.

Achievements And Certificates

Hackathon Achievement:

- HackRush (AIML): Secured 1st Place at Coimbatore Institute Of Technology. *March 2025*
- Caterpillar Tech Challenge 2026: Top 10 National Finalist among 1,500+ teams. *June 2026*

Course Certifications:

- Stanford ML
- Deep Learning
- Gen AI (Google Cloud)
- AI (IBM)
- Cloud Computing
- Python (Google)

Areas of Interests

Data Science, Machine Learning, Deep Learning, Computer Vision, Generative AI, Agentic AI.

Education

- **M.Sc - Artificial Intelligence and Machine Learning (Integrated)** *2022-2027*
  Coimbatore Institute of Technology, Coimbatore, India. CGPA: 7.1

Research and Presentations

- **Predicting the Attacks using AI in MANETs,** Coimbatore Institute of Technology. *May 2024*
- **Industry 5.0 and Cloud Security,** Coimbatore Institute of Technology. *October 2024*